

RES-5 2PI resummation strategy: the repulsive screening tames the marginal

TECT verification-first repository · claim B1-RH-ENUM

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1. Purpose and scope

The dressed-skeleton note found the loop parameter $g = \lambda' B_d = 1.03 = O(1)$ marginal, so the skeleton series does not geometrically decay – a resummation is required. The operator named three routes (A: 2PI self-consistent; B: Borel / optimal truncation; C: sign / cancellation). This note executes the strategy and finds that Routes A and C COINCIDE here: the repulsive dressed vertex supplies the screening sign that makes the 2PI/Dyson resummation finite. The result is a viable closure route, at strong-evidence grade; the rigorous 2PI computation is the residual. Scope: the screening mechanism and the difference scale; anchor μ^2 .

2. The marginal series is divergent, but the resummation is finite

The dressed quartic is REPULSIVE: the bare $u = -0.86 < 0$ (attractive, Brazovskii) is lifted by the sextic dressing to $\lambda' = 3u_{\text{eff}} = 8.06 > 0$. The RPA/Dyson density susceptibility therefore carries the SCREENING sign,

$$\chi = \chi_0(1 - g + g^2 - \dots) = \frac{\chi_0}{1 + g}, \quad g = \lambda' B_d = 1.03.$$

Term-by-term the series diverges ($|g| > 1$; the partial sums grow), but the 2PI/Dyson SELF-CONSISTENT solution is finite:

$$\frac{\chi_{\text{resummed}}}{\chi_0} = \frac{1}{1 + g} = 0.49 \quad (\text{screened} \sim \text{half}).$$

The divergent perturbative series is merely the asymptotic expansion of this finite self-consistent free energy, which is the physical value. (Contrast: an ATTRACTIVE vertex would give $1/(1 - g)$ with $g > 1$ – a divergence / instability. The repulsive screening sign is what makes the vacuum stable AND the resummation finite: the same fact.)

3. The resummed difference is finite and a_0 -suppressed

The 2PI self-consistent dressing is pattern-independent at fixed I (the common sea), so the common-mode part cancels in the selection difference and the pattern-dependent part is carried by a_0 :

$$\Delta F_{\text{exact}} \sim a_0 \chi_{\text{resummed}}^{\text{pd}} \sim a_0 \cdot 0.49 \cdot (\dots) \sim 0.047 \cdot (\dots),$$

FINITE (screened) and a_0 -suppressed. The selection survives the resummed exact free energy: the marginal series does not threaten the sign, because its physical (resummed) value is the screened, a_0 -suppressed difference, not the divergent term-by-term sum.

4. Status and the residual

T4 STRONG EVIDENCE (for the closure ROUTE). Established: the dressed vertex is repulsive (screening sign $1/(1+g)$); the marginal/divergent series resums (2PI/Dyson) to a finite screened susceptibility $0.49\chi_0$; the common-mode $+a_0$ structure makes the resummed difference finite and a_0 -suppressed (~ 0.047). **OPEN (the residual):** the rigorous 2PI self-consistent computation of ΔF_{exact} and the quantitative bound $|\Delta F_{\text{exact}}| < \Delta F_{\text{margin}}$ – i.e. verifying the common-mode cancellation at the self-consistent level and computing the screened pattern-dependent susceptibility explicitly. No tier flip: B1-RH-ENUM T6 CONDITIONAL on {H-LAYER}. **B1 closure** $\iff$ this 2PI self-consistent bound, for which the screening route is now established.

5. Devil’s-advocate / self-adversarial review (CLAUDE.md 6.3)

- (α) “ $1/(1+g)$ with $g > 1$ sums a divergent series – is it physical?” VALID-with-mitigation: the physical object is the 2PI/Dyson SELF-CONSISTENT solution (obtained by solving the Dyson equation, not summing the series), which is manifestly finite; the divergent series is its asymptotic expansion. This is standard self-consistent field theory, not an illegitimate resummation of a divergence.
- (β) “ $0.49\chi_0$ is the RPA bubble-chain only; vertex corrections could change it.” VALID: the RPA/2PI bubble chain is the leading self-consistent resummation; vertex corrections are higher 2PI skeletons, the residual. But the SIGN (screening, finite) is robust – it follows from $u_{\text{eff}} > 0$, not from the truncation.
- (γ) “The difference ~ 0.047 omits the $(\dots)$ factor.” VALID: the $a_0 \cdot 0.49$ is the SCALE; the precise resummed pattern-dependent susceptibility (the $(\dots)$) is the residual 2PI computation. The strong evidence is the route (screening + common-mode + a_0), not a certified number.

Quantitative sanity check (mandatory): *sign* – $u_{\text{eff}} > 0$ gives the screening denominator $1+g$, not the instability $1-g$; *finiteness* – $1/(1+1.03) = 0.49$ (finite despite $g > 1$); *divergence* – the partial sums $1, -0.03, 1.03, \dots$ grow (the series is asymptotic, the 2PI solution is the value); *scale* – $a_0 \cdot 0.49 \approx 0.047$ (a_0 -suppressed, screened).

6. Result footer

Result ID:	B1-RH-ENUM / RES-5 2PI resummation strategy (screening)
Precise statement:	<code>g = lam' B_d = 1.03 > 1 (marginal/divergent series), but u_eff>0 (repulsive; bare u=-0.86 lifted by the sextic) => RPA/Dyson screening sign 1/(1+g): the 2PI self-consistent susceptibility is FINITE, chi_resummed = chi_0/(1+g) = 0.49 chi_0. The divergent series is its asymptotic expansion. Common-mode cancellation + a0 => the resummed pattern-dependent DIFFERENCE ~ a0*0.49 ~ 0.047, finite and a0-suppressed -- the selection survives the resummed exact free energy. RES-5 closure ROUTE established; rigorous 2PI computation the residual.</code>
Scope:	Screening mechanism + difference scale; anchor μ^2 . The rigorous 2PI self-consistent bound is OPEN.
Dependencies:	<code>res5-dressed-threeloop-skeleton-bound (g=1.03 marginal); hdiag-fullcovariance-formulation (u_eff>0 dressing); res5-commonmode-envelope (a0); Math424-AddA.</code>
Evidence grade:	ANALYTIC + EXECUTED (res5_rpa_screening.py 5/5).

Reproduction command: `python codes/vacuum/res5_rpa_screening.py`
 Expected output: `u_eff>0` (repulsive); `1/(1+g)=0.49` finite; series divergent but resummation finite; `a0*0.49~0.047`; 5/5 PASS.
 Falsification gate: `u_eff<=0` at the anchor (would give the instability sign `1/(1-g)`); or a 2PI computation giving `|Delta F_exact| >= Delta F_margin` (resummed difference exceeds the margin).
 Tier before / after: No tier flip. B1 T6 on {H-LAYER}. RES-5: closure ROUTE via 2PI screening established (T4); rigorous 2PI bound the residual.
 No-overclaim statement: This does NOT close RES-5. It establishes the resummation ROUTE (repulsive screening makes the marginal series resum to a finite, `a0`-suppressed difference) at strong-evidence grade; the rigorous 2PI self-consistent computation and the quantitative margin bound are the residual. B1 stays T6 on {H-LAYER}.
 Next required action: Carry the 2PI self-consistent computation of the screened pattern-dependent susceptibility and verify `|Delta F_exact| < Delta F_margin`; then RES-5 $\rightarrow$ controlled-error theorem and H-LAYER's last axis closes.